Vishrut Pankaj Chokshi

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Professional Summary

Process and fabrication engineer with hands-on cleanroom experience spanning thin-film deposition, photolithography, plasma etching, and dielectric characterization. Skilled in DOE, SPC, and failure analysis (FE-SEM/EIS). Six Sigma Yellow Belt | MS in Electrical and Computer Engineering.

Education

Michigan Technological University — Aug 2024 – Dec 2025
MS in Electrical and Computer Engineering — GPA: 3.59/4.0
Graduate Certificate in Electronics and Materials Processing

Sardar Patel Institute of Technology — Aug 2019 – May 2023
B.Tech in Electronics and Telecommunication Engineering — CGPA: 8.72/10.0

Work Experience

Graduate Research Assistant, Michigan Technological University – Houghton, MI — Feb 2026 – Present
Research conducted since Sept 2025; official appointment upon OPT approval (Feb 2026)

- Developed a non-destructive screening workflow combining FE-SEM imaging and Electrochemical Impedance Spectroscopy (EIS) to detect and characterize nanoscale pinholes in sub-35 nm SiO_2 dielectric thin films.
- Authored and updated Standard Operating Procedures (SOPs) for cleanroom instrumentation, ensuring consistent execution of deposition, imaging, and impedance measurement protocols across users.
- Assisted the Lab Manager in troubleshooting and restoring operability of cleanroom tools, supporting preventive maintenance activities to minimize downtime and maintain process continuity.

Research and Development Intern, EdgeLabs LLP – Mumbai, India — Feb 2023 – Jun 2023

- Verified gold nanoparticle size using optical spectroscopy and used the nanoparticles in rapid detection kits for four targets (syphilis, hepatitis B, dengue, malaria); evaluated kit performance by inspecting test and control line response against expected outcomes.
- Reviewed and refined process protocols for a Troponin I rapid detection kit by analyzing parameter deviations; contributed to protocol adjustments that improved test results.

Relevant Experience

Test Chip Fabrication and Packaging, Michigan Technological University — Jan 2025 – Apr 2025

- Fabricated and packaged a test chip through wafer-level processes, including PVD sputtering (30nm Cr and 300nm Au at 100W RF), plasma etching, photolithography (coating of ~1.5µm SU-8 photoresist), and wafer dicing.
- Tuned tool setup and parameters (RF sputterer, etcher, wire bonder) while applying DOE-based optimization of lithography with 12–14s exposure and 35–50s development, and bonding processes to improve consistency under faculty supervision.
- Designed custom 2in×2in, 2-layer PCB interposer (10mil traces, 30/15mil vias) for system-level testing using KLayout and Altium, integrating 500mil×500mil BGAs with vendor and in-house fabrication constraints for assembly compatibility.
- Supported chip assembly, including die attach of test dies to QFN72 packages (epoxy cure at 150°C for 15min), electrical validation of resistor structures, and SOP documentation to improve reproducibility and process steps to device outcomes.

Integrated NMOS Transistor Fabrication, Michigan Technological University — Sept 2024 – Dec 2024

- Fabricated NMOS transistors using two-step oxidation, diffusion, photolithography, metallization (50nm Cr / 200nm Au), WSi_2 gate deposition, and dicing into 2mm×6mm dies under ISO cleanroom protocols.
- Developed patterns through photolithography for W ≈ 10µm / L ≈ 2µm devices by tuning spin, exposure dose (9–15mW/cm^2 intensity), while coordinating PVD and diffusion splits to improve feature resolution, accuracy, and yield.
- Extracted threshold voltage (V_{th}), on/off current ratio, and subthreshold swing from IV measurements; verified performance against design targets and identified process-limiting steps across splits.
- Root cause analysis of gate oxide non-uniformities and lithographic misalignment; adjustments to spin speed and exposure dose improved feature resolution and cross-wafer consistency.

Skills

Equipment: RF PVD Sputterer, Plasma Etcher, Optical Lithography (UV), Wire Bonder, ThermoFisher Apreo2 FE-SEM
Characterization: EIS, IV/CV, SEM/EDS, Profilometry, Spectroscopy
Methods: DOE, SPC, DMAIC, Root Cause Analysis, Failure Analysis
Software: MATLAB, Python